



DEPARTMENT OF Electrical ENGINEERING **COLLEGE OF E&ME, NUST, RAWALPINDI**

EE-211 Electrical Network Analysis:

Project Report:

Course Instructor: Dr. Zubair Ahmed

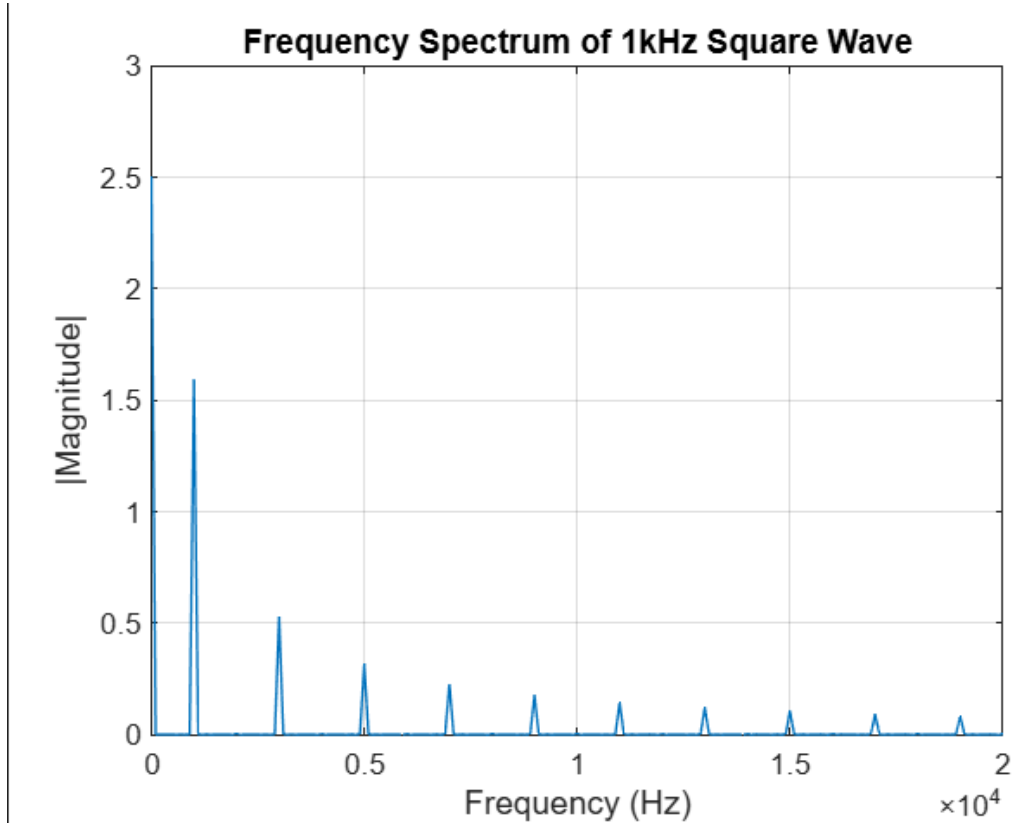
Lab Instructor: LE. Aleena

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Outputs Required

Output no:	Waveform:	Achieved:
1	Extract the DC component from the square input waveform: DC output level $2.5V \pm 10mV$	
2	Extract the 1kHz sinusoidal signal from the square input	
3	Extract the 5kHz sinusoidal signal from the square input	
4	Combine the 1kHz fundamental and 5kHz harmonic signal to get a 5kHz signal modulated by 1kHz signal.	
5	Remove the DC component from the original square waveform using a passive circuit, resulting in a zero-centered AC signal.	

FFT of Input:



FILTER 1: Low-pass Filter:

Description:

- A **passive RC low-pass filter**, implemented as a cascaded 2-stage design.

Analysis:

1. Filter Type:

Low-pass

2. Filter Order:

2nd order (each RC stage contributes a 1st order roll-off)

3. Topology:

Cascaded RC filters

4. Cutoff Frequency:

$$f_c = \frac{1}{2\pi RC}$$

$$f_c = \frac{1}{2\pi \cdot 1k \cdot 10u}$$

$$f_c = 15.9Hz$$

5. Justification:

- The filter is designed to pass very low-frequency components (DC or near-DC) and suppress higher frequencies.
- A 2-stage RC provides a sharper roll-off than a single stage.

FILTER 2: LC Low-pass Matching Filter:**Description:**

- A **4-stage LC low-pass filter** with 50 Ω terminations, suggesting impedance matching (possibly for RF applications).

Analysis:**1. Filter Type:**

Low-pass

2. Filter Order:

8th order (4 L-C π -sections)

3. Topology:

Ladder (π -section)

4. Cutoff Frequency:

$$f_c = \frac{1}{2\pi\sqrt{LC}}$$

$$f_c = \frac{1}{2\pi\sqrt{10mH \cdot 100nF}}$$

$$f_c = 5.03kHz$$

5. Justification:

- Provides steep attenuation above 5 kHz.
- 50 Ω terminations are standard in communication systems.
- Passive design ensures wide bandwidth and linear phase in passband.

FILTER 3: LCLCRC Low-pass Filter:

Description:

- A more complex **passive LC-RC filter**, combining damping resistors for stability.

Analysis:

1. Filter Type:

Low-pass

2. Order:

Approximately 4th order (2 inductors + 2 capacitors)

3. Topology:

Ladder filter with damping resistors

4. Cutoff Frequency:

$$f_c = 503\text{Hz}$$

5. Justification:

- Damping resistors (10 Ω) prevent peaking/resonance.
- Targets sub-kHz low-pass filtering.
- Passive for simplicity and robustness.

FILTER 4: Active Amplifier Stage:

Description:

- Operational amplifier configuration (likely **non-inverting amplifier** with gain).

Analysis:

1. Purpose:

Amplification of filtered signal (possibly buffering)

2. Gain:

$$A_v = 11$$

3. Justification:

- Used to boost the signal level from passive filter stage.
- Op-amp ensures low output impedance for driving the next stage.

FILTER 5: LC Low-pass Filter:

Description:

- Another **passive low-pass filter**, focused on higher frequency applications.

Analysis:

1. Filter Type:

Low-pass

2. Order:

6th order (3 L-C sections)

3. Cutoff Frequency:

$$f_c = 5.03kHz$$

4. Justification:

- Designed to suppress high-frequency content from the amplifier.
- Series inductors + shunt capacitors provide steep roll-off.

FILTER 6: Capacitive Tap:

Description:

- A capacitive coupling and load.

Analysis:

1. Purpose:

Possibly sampling high-frequency content or DC blocking.

2. RC time constant:

$$\tau = 5ms$$

$$f_c = 31.8Hz$$

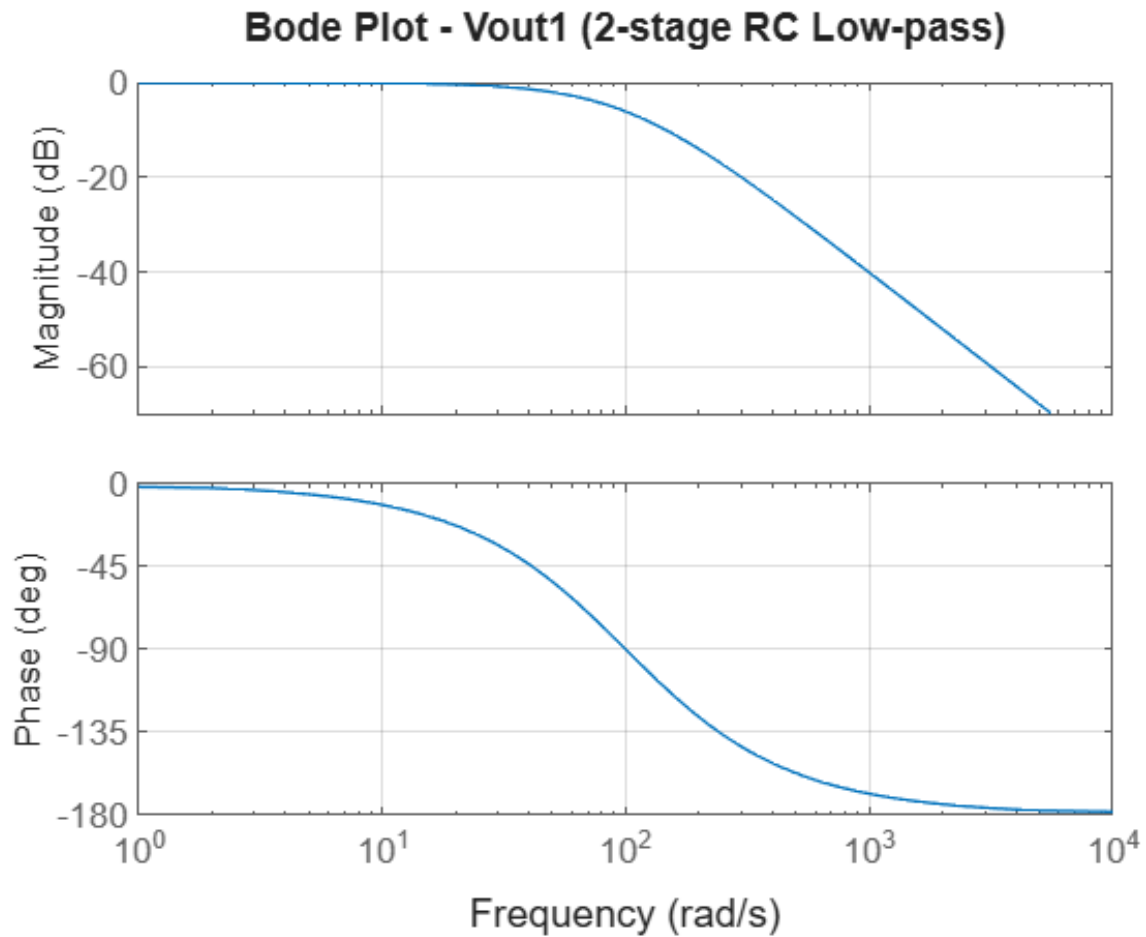
3. Justification:

- Blocks DC, samples AC component.
- Light load due to high R value.

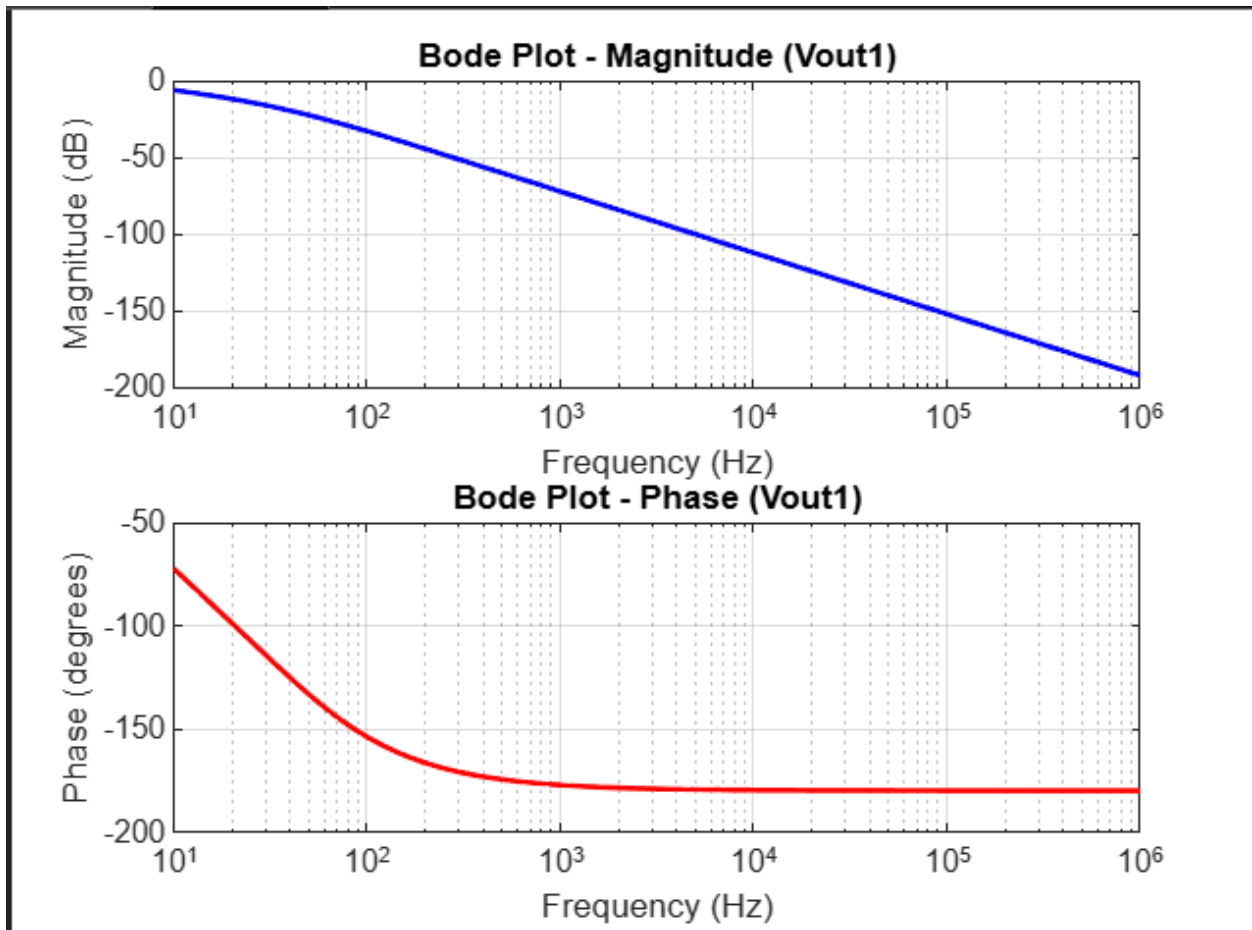
Comparing Bode Plots:

Output 1:

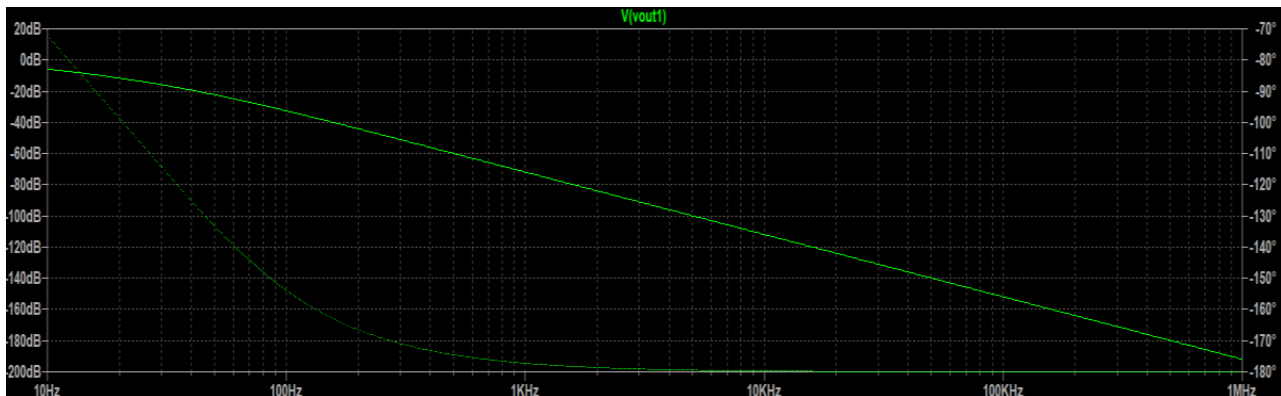
Theoretical:



Measured:

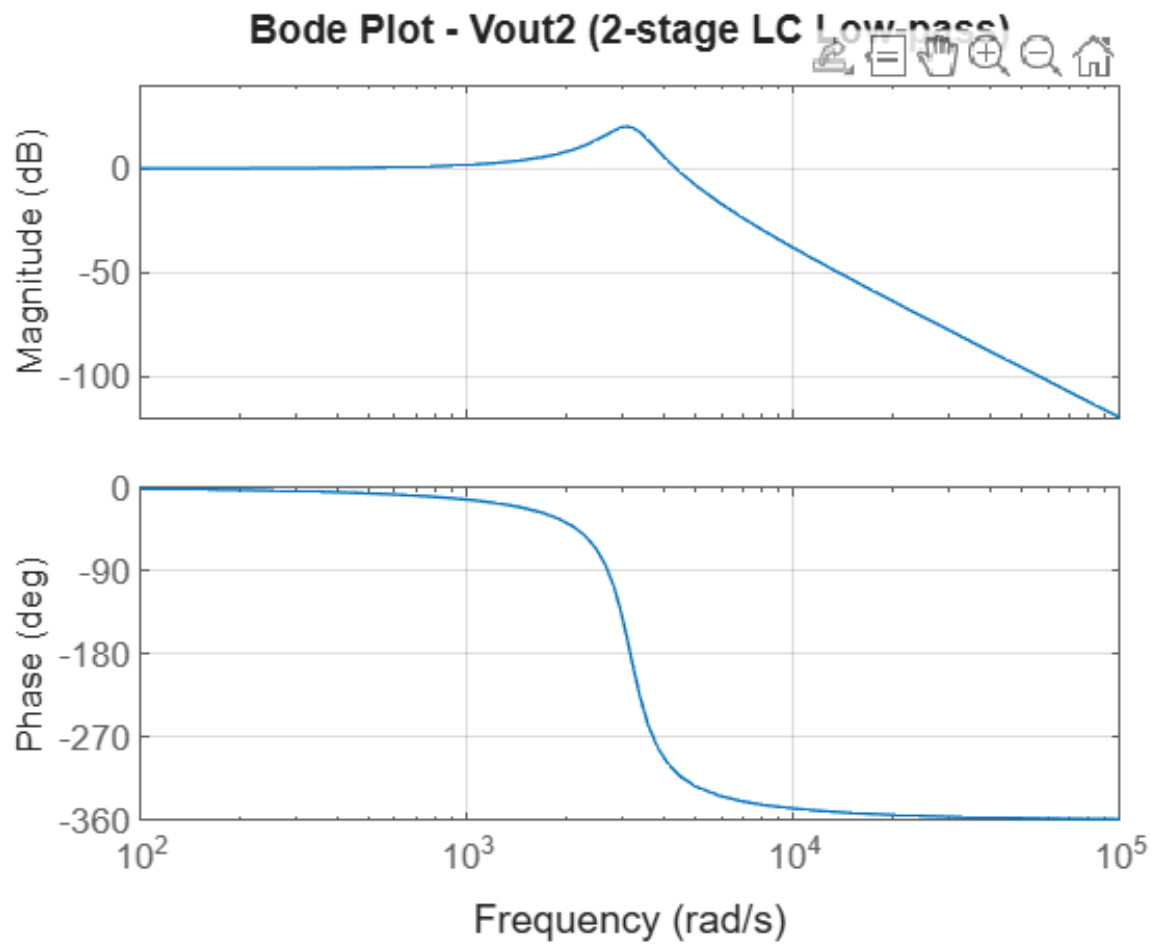


Simulated:

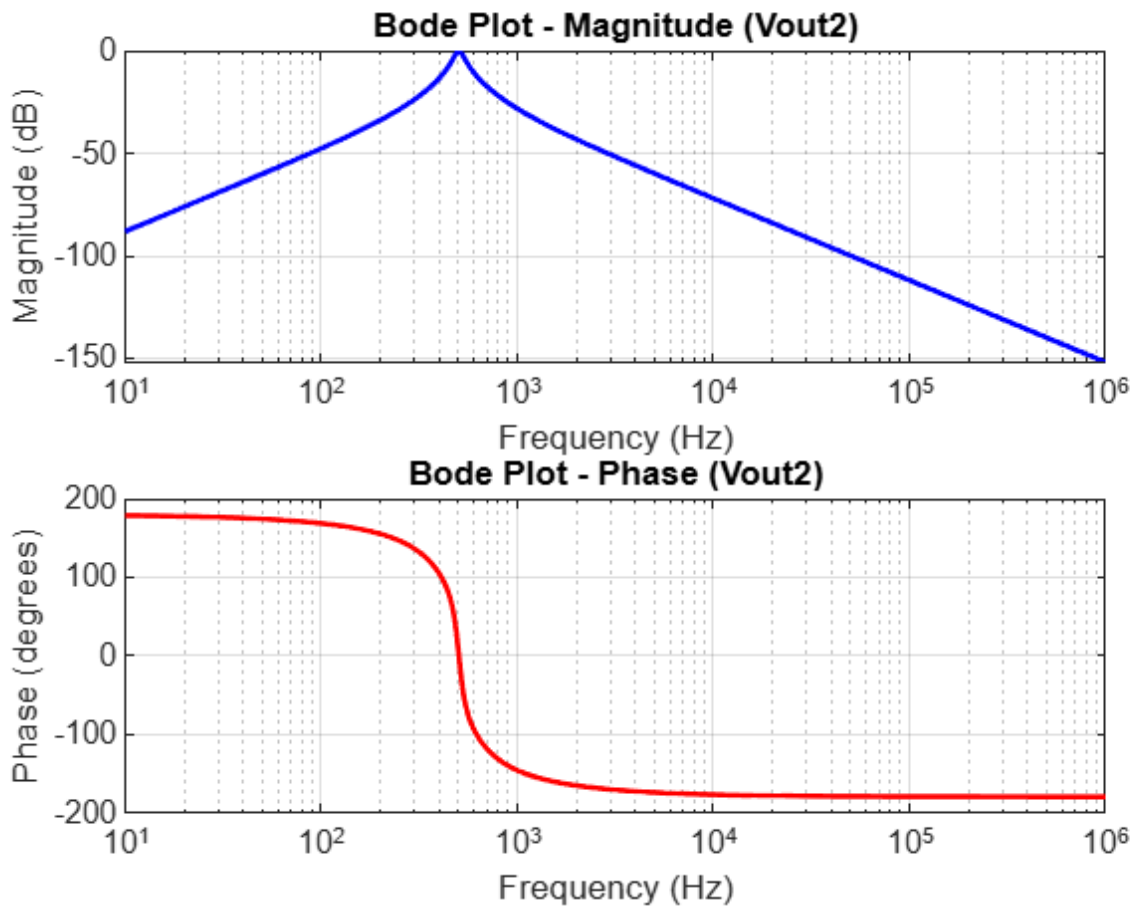


Output 2:

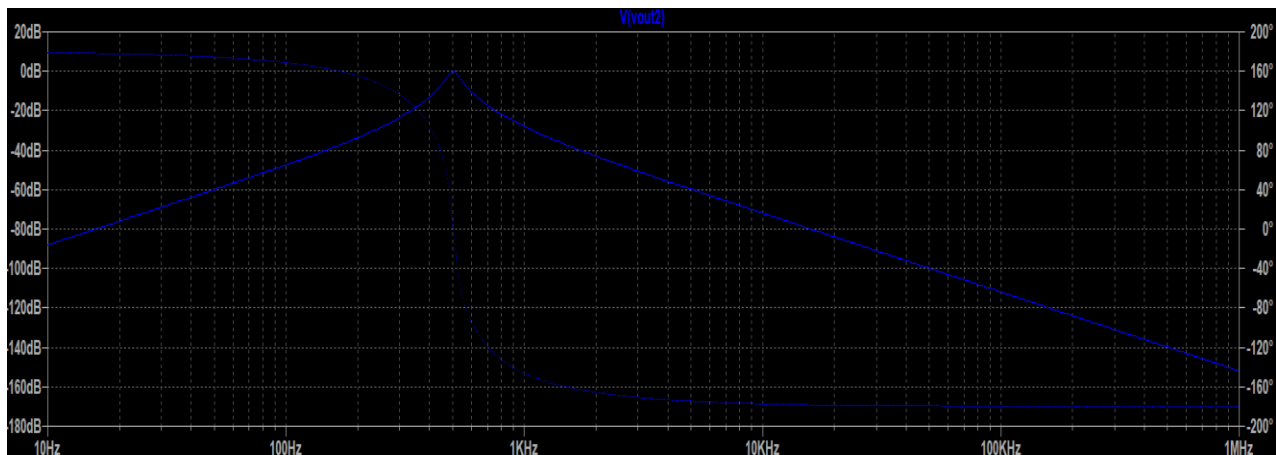
Theoretical:



Measured:

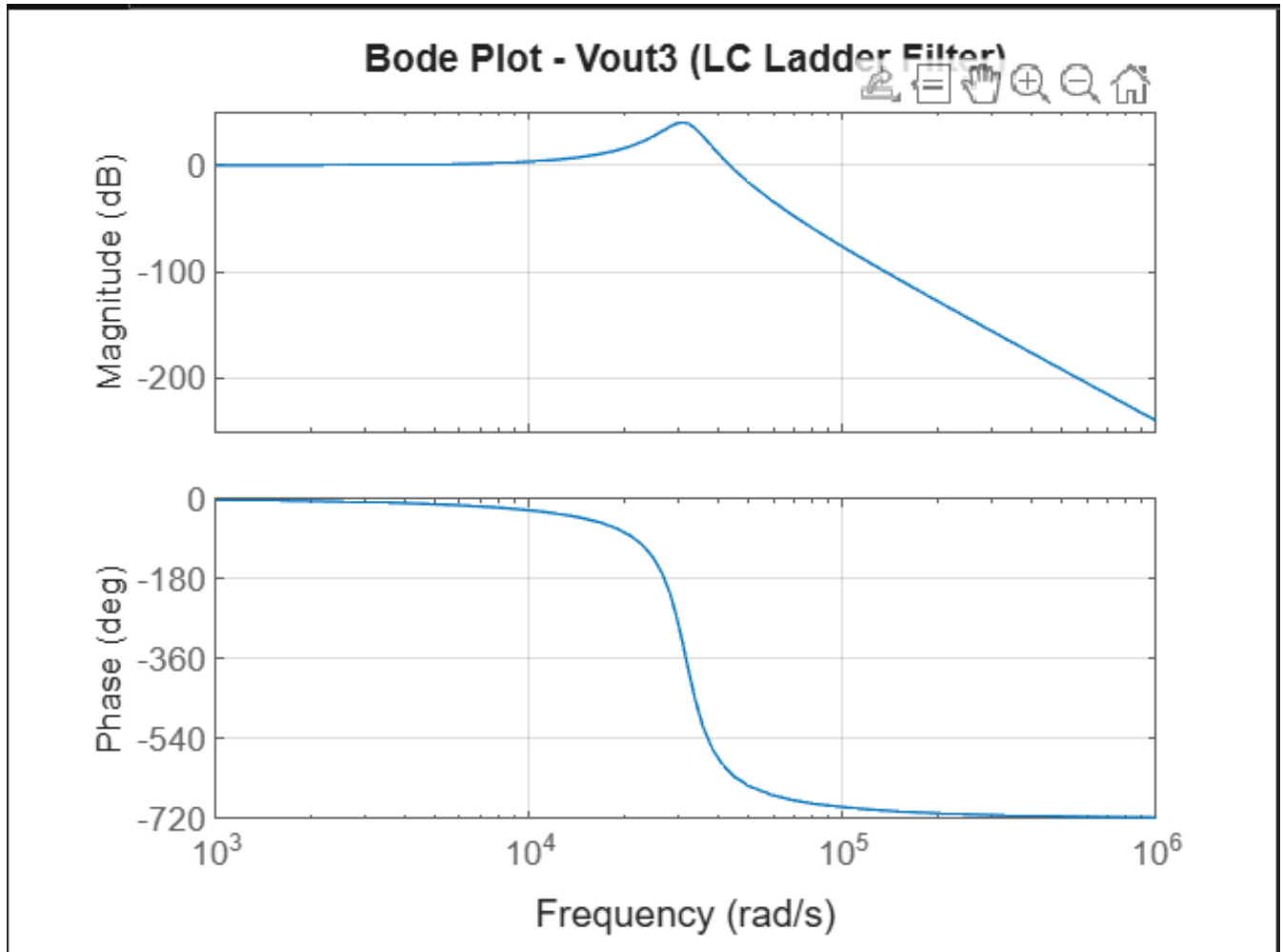


Simulated:

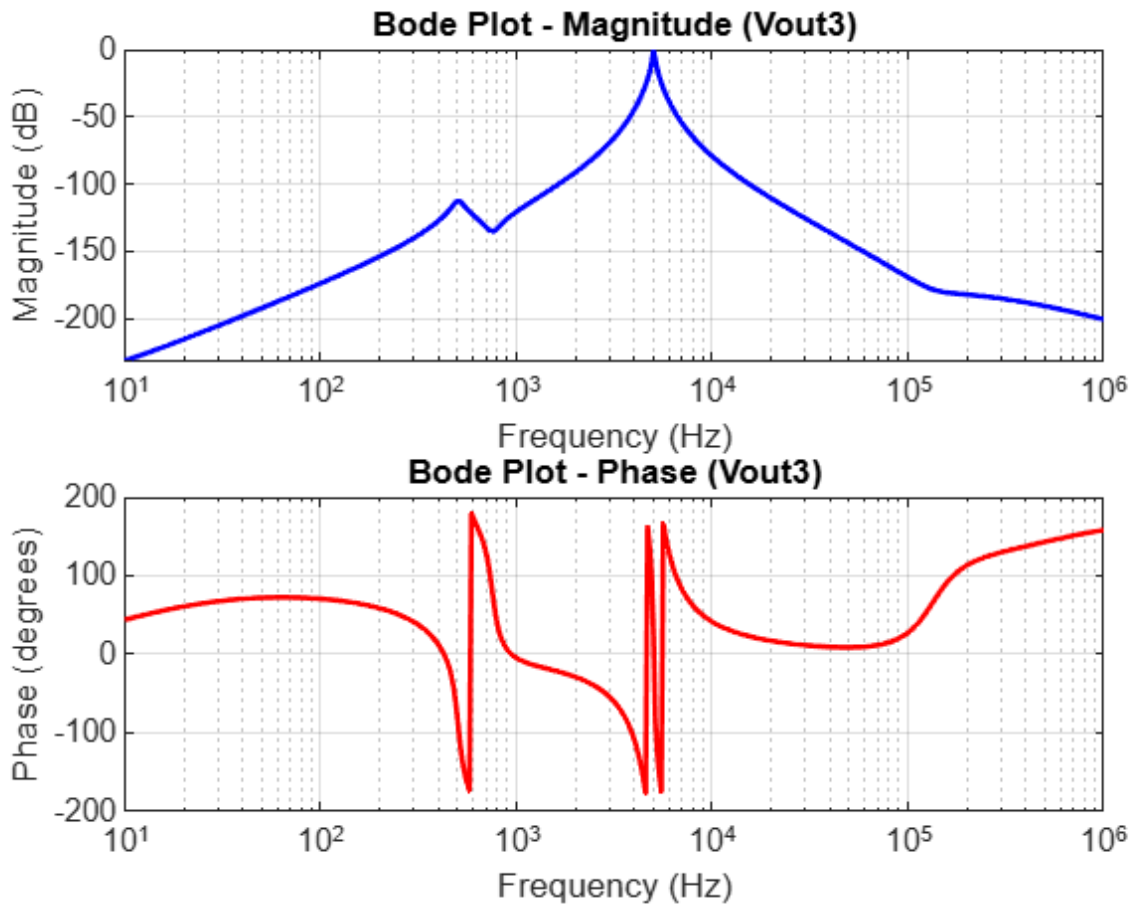


Output 3:

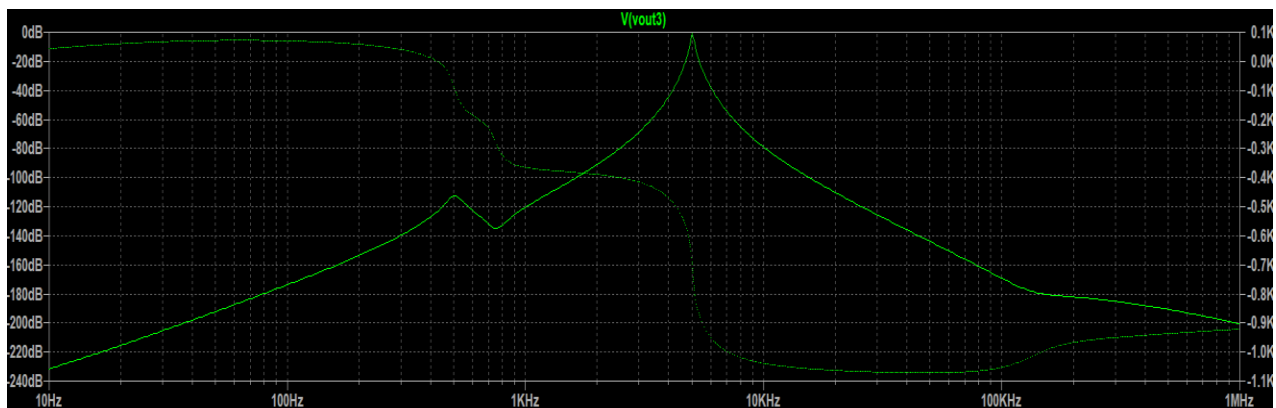
Theoretical:



Measured:

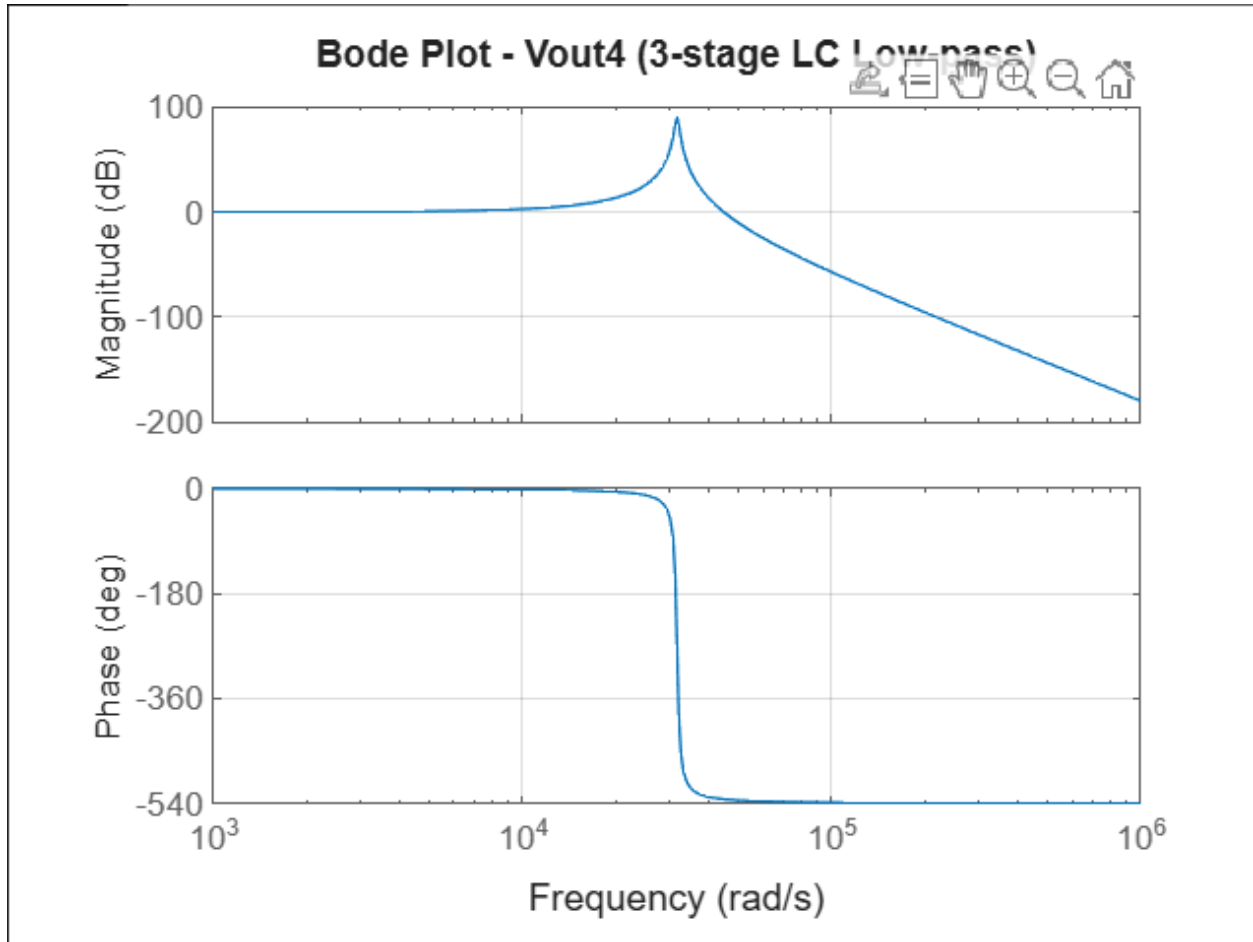


Simulated:

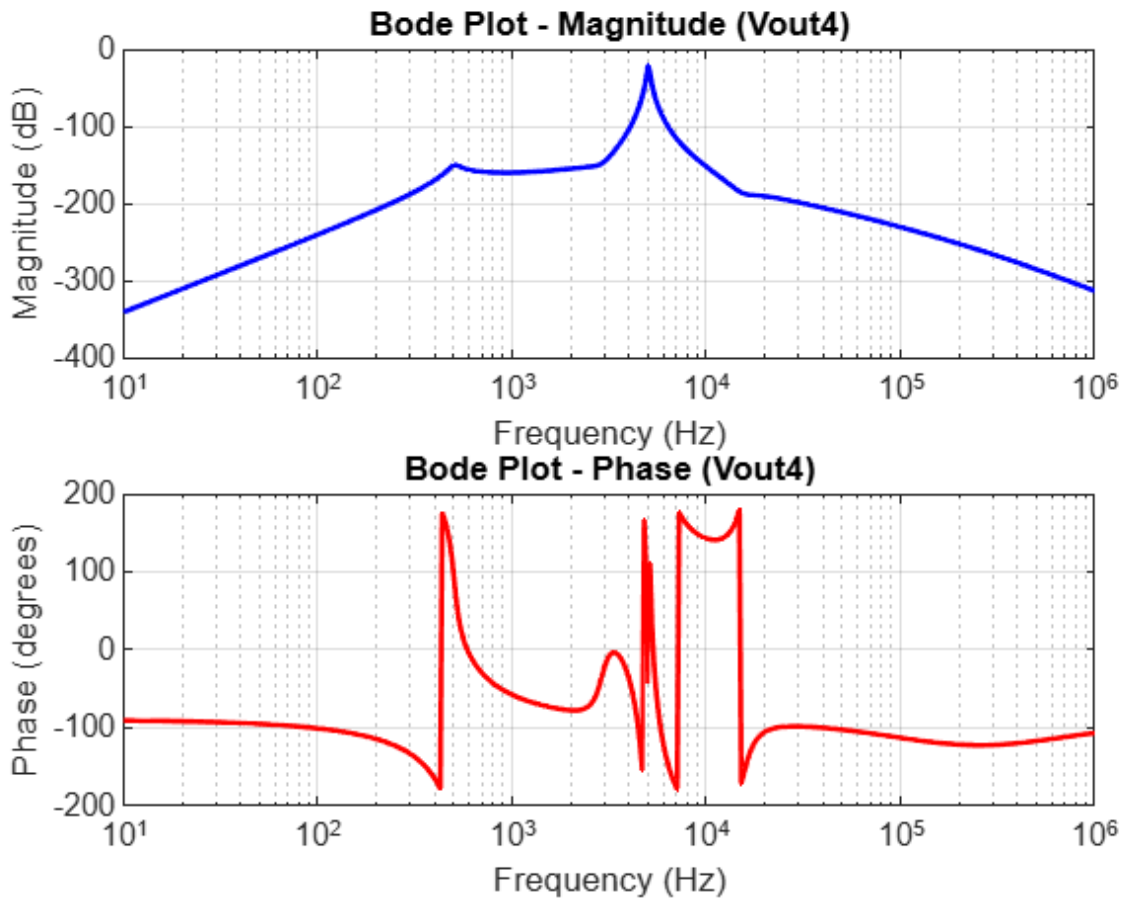


Output 4:

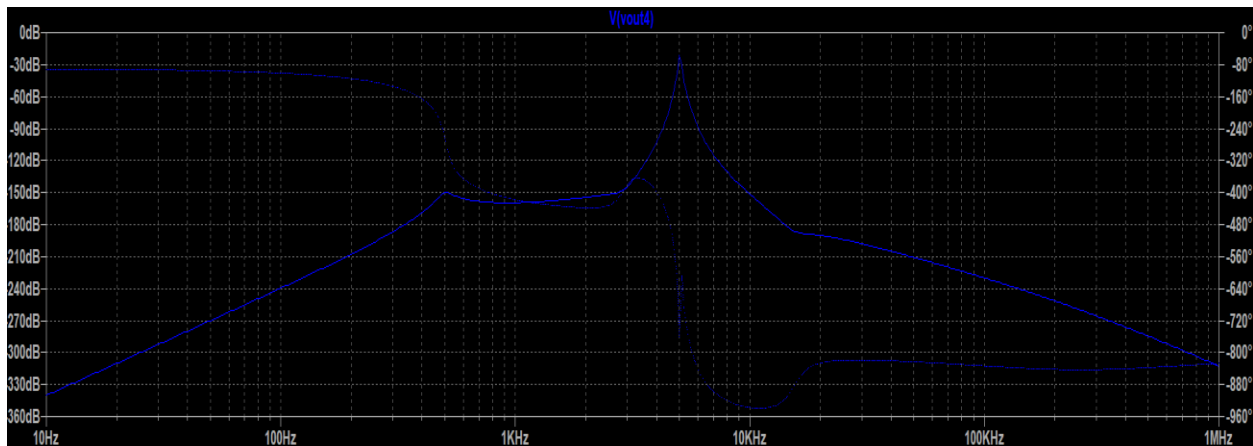
Theoretical:



Measured:

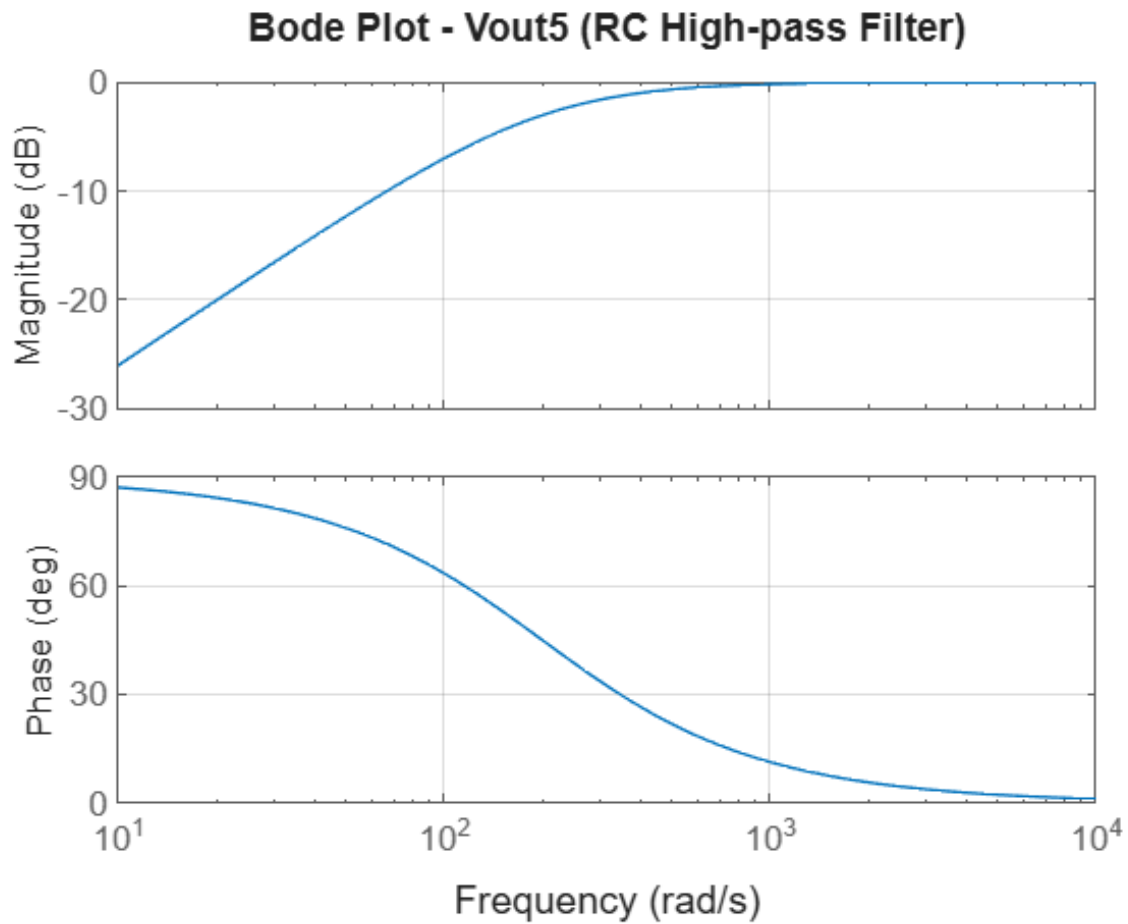


Simulated:

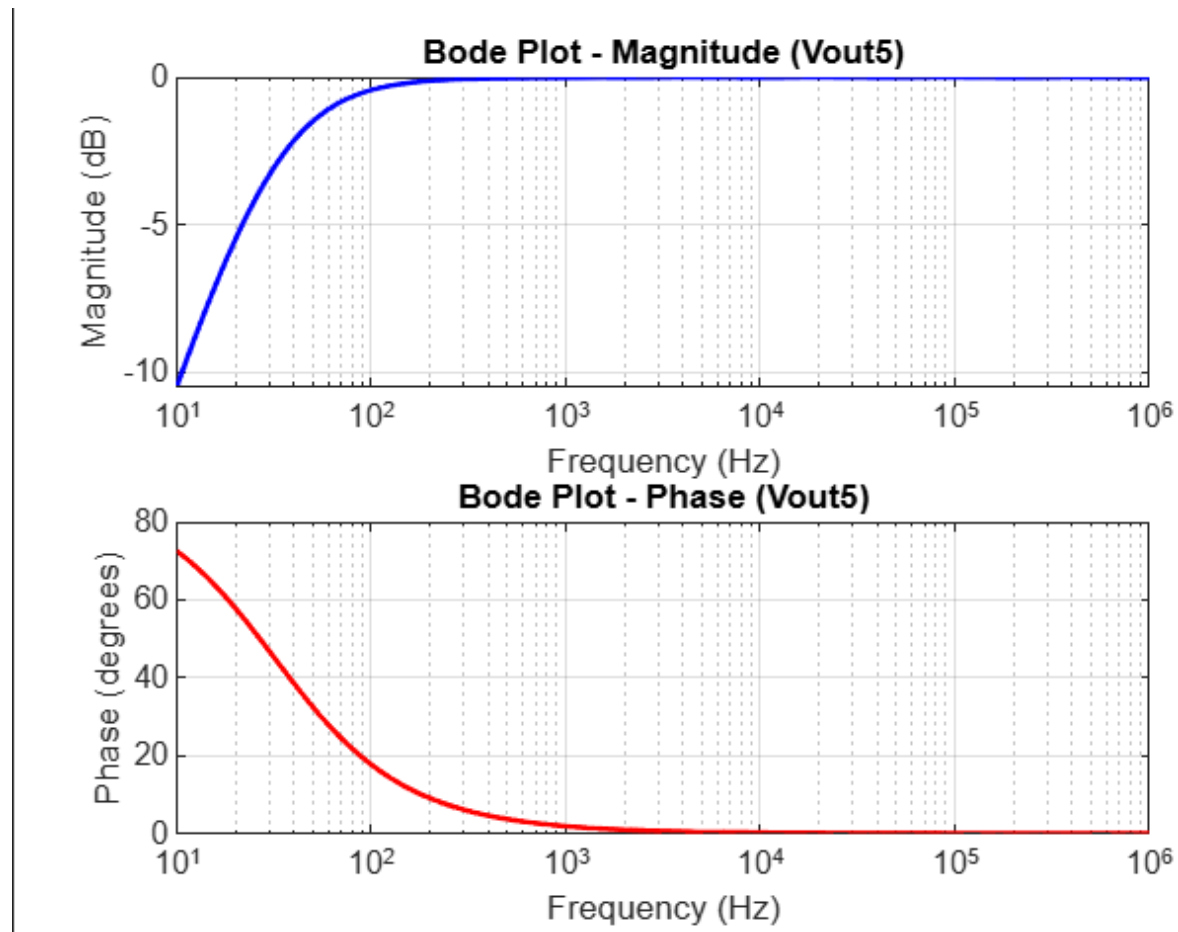


Output 5:

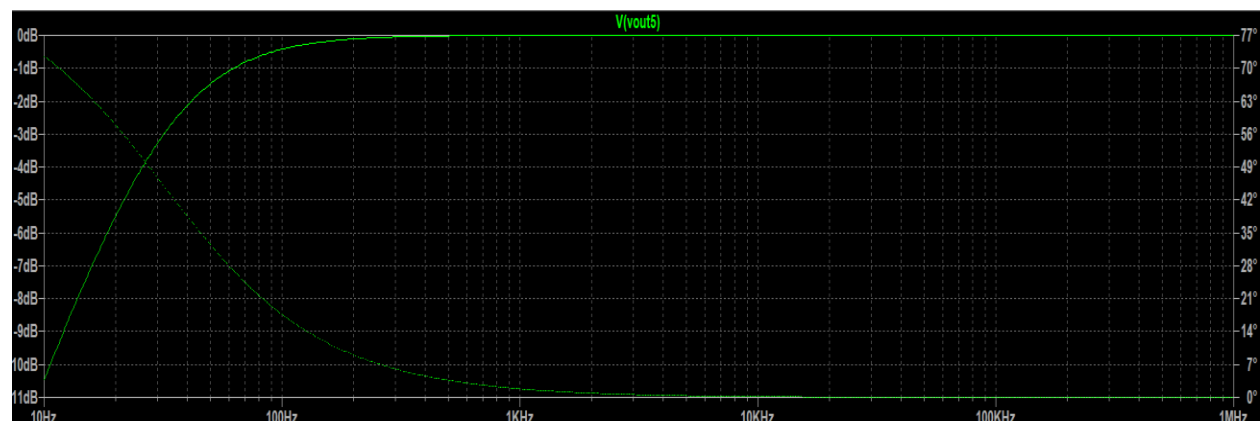
Theoretical:



Measured:

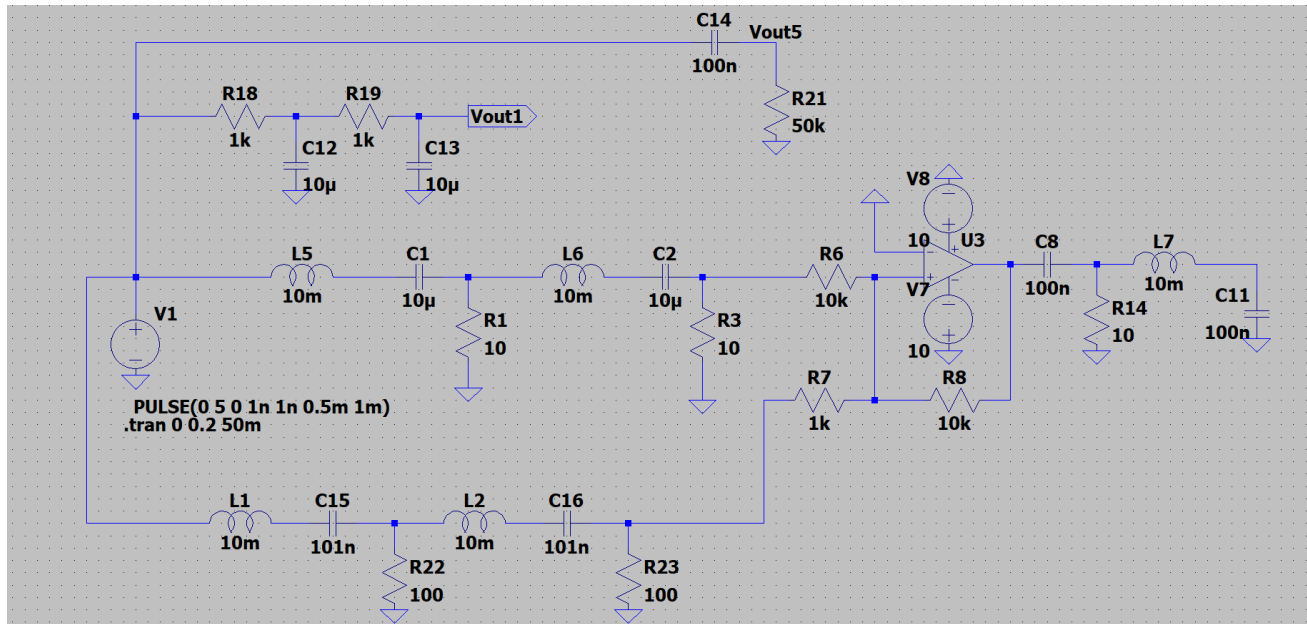


Simulated:

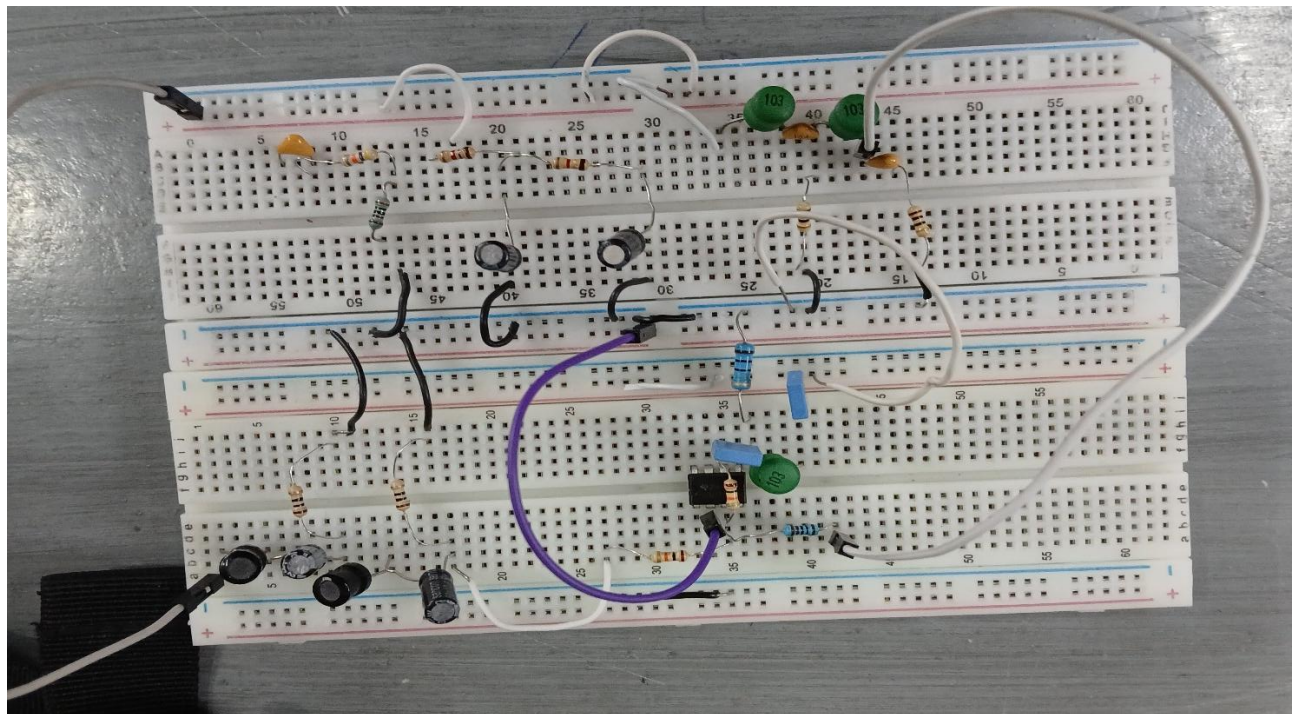


Comparing Graphs

Simulation:

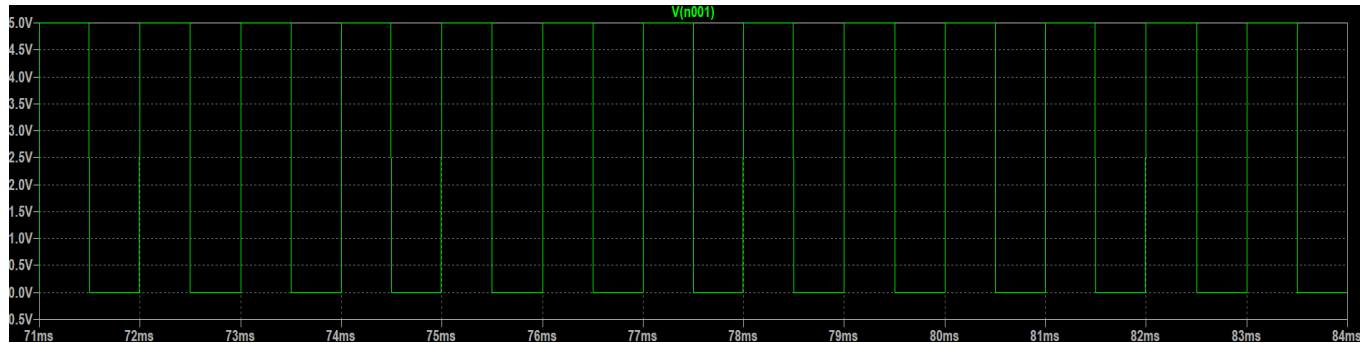


Implemented:

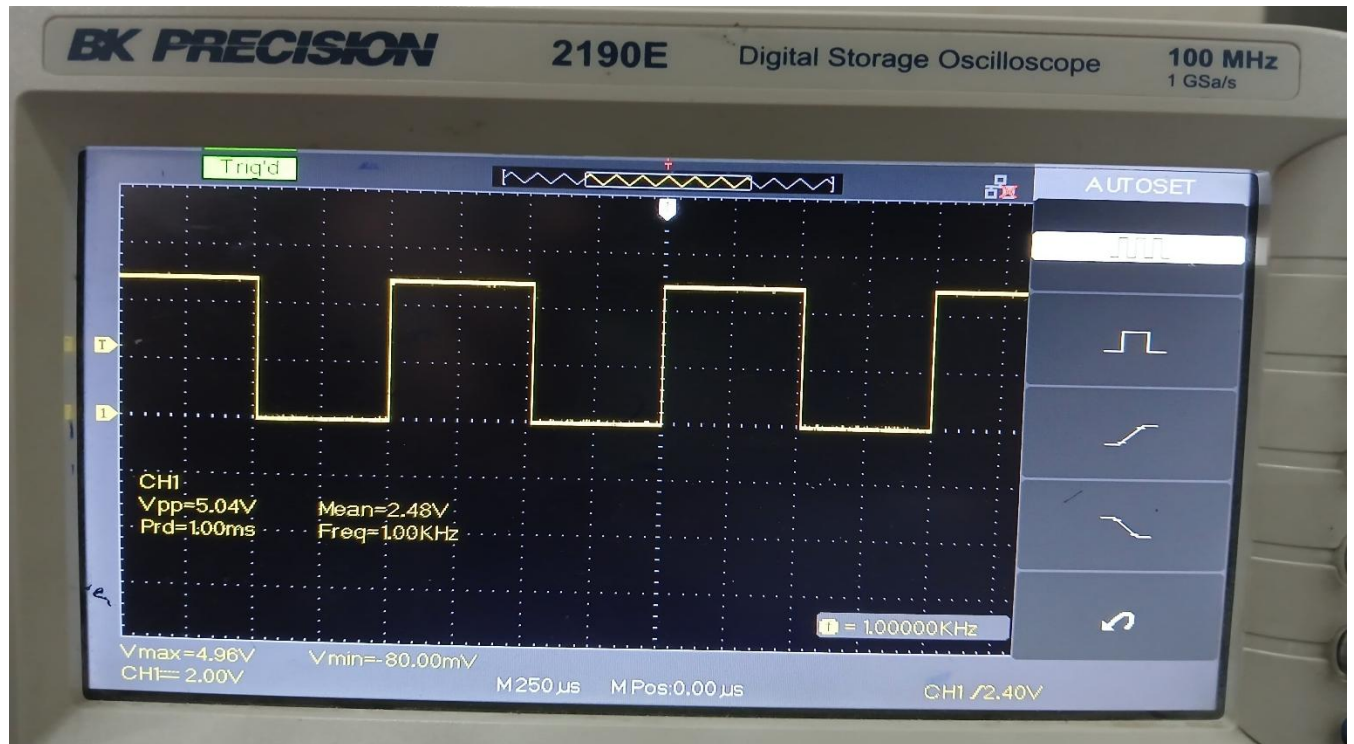


Input:

Simulated:

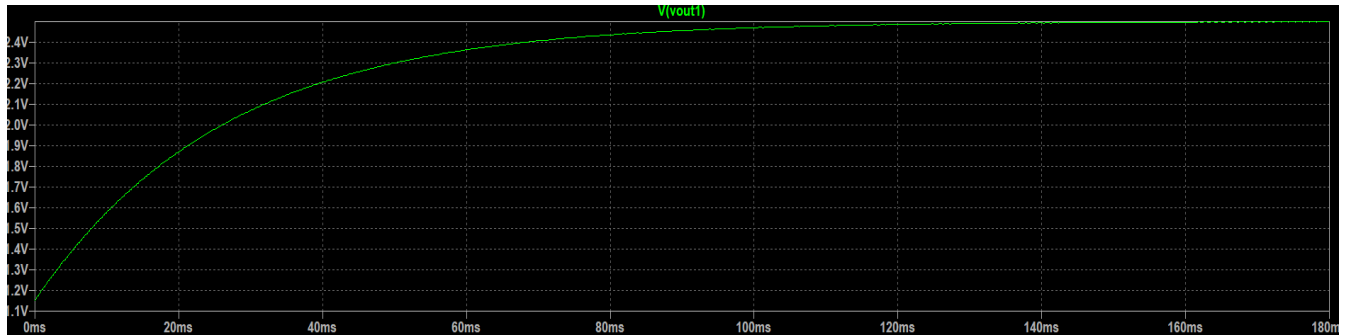


Measured:

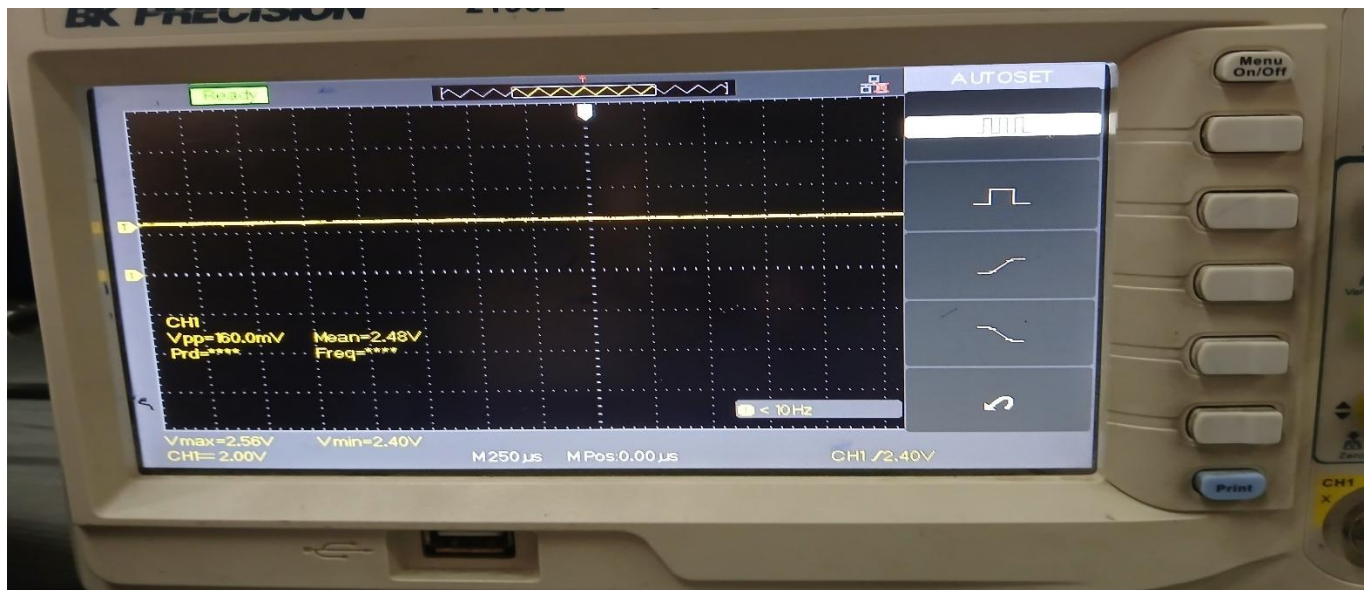


Output 1:

Simulated:

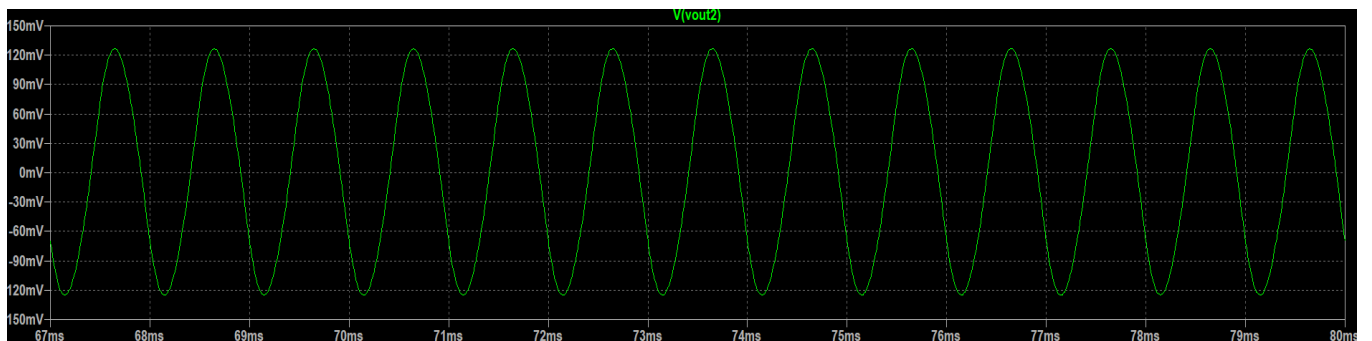


Measured:

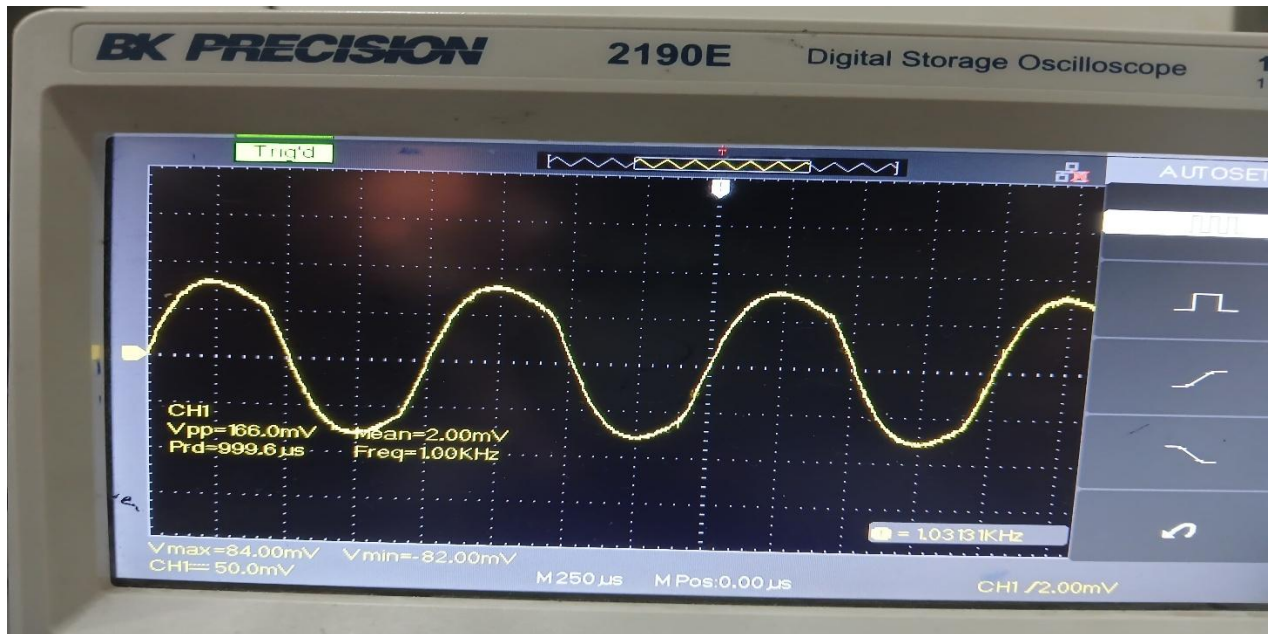


Output 2:

Simulated:

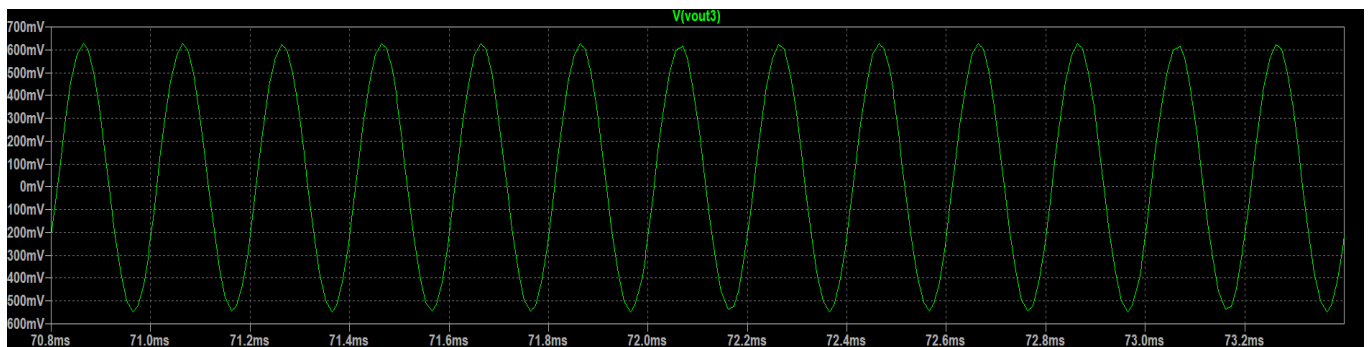


Measured:

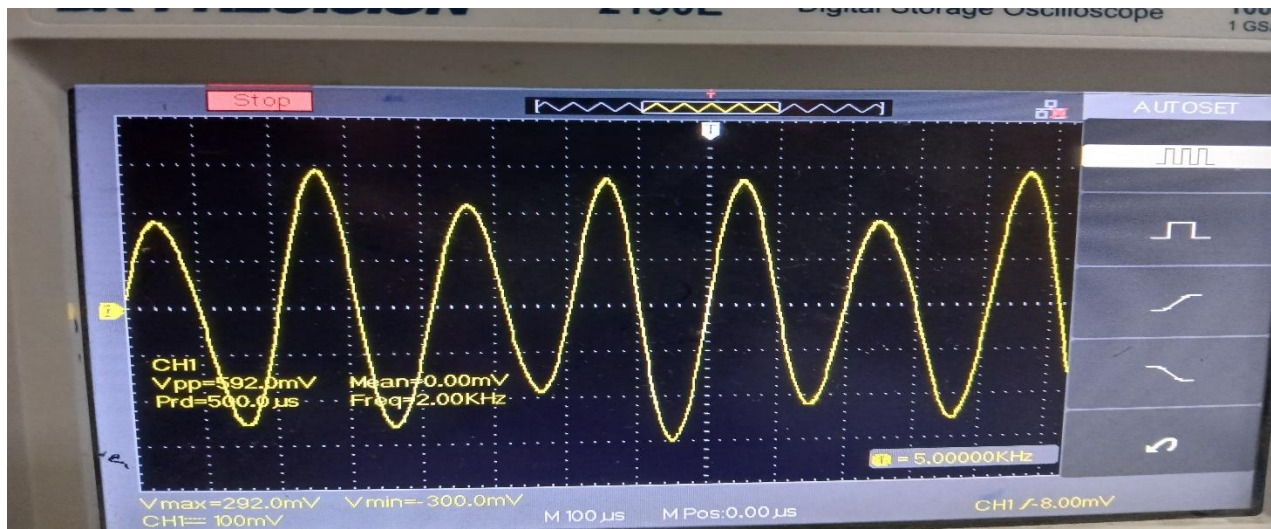


Output 3:

Simulated:

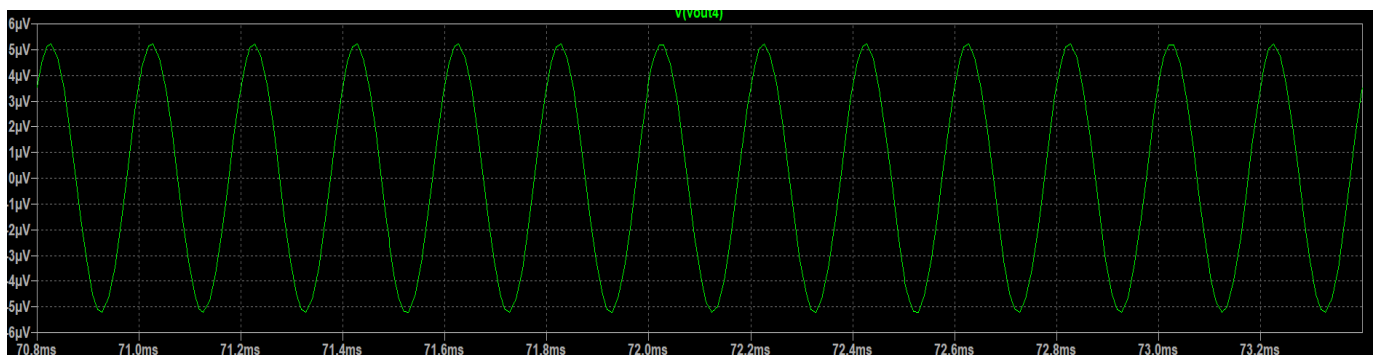


Measured:

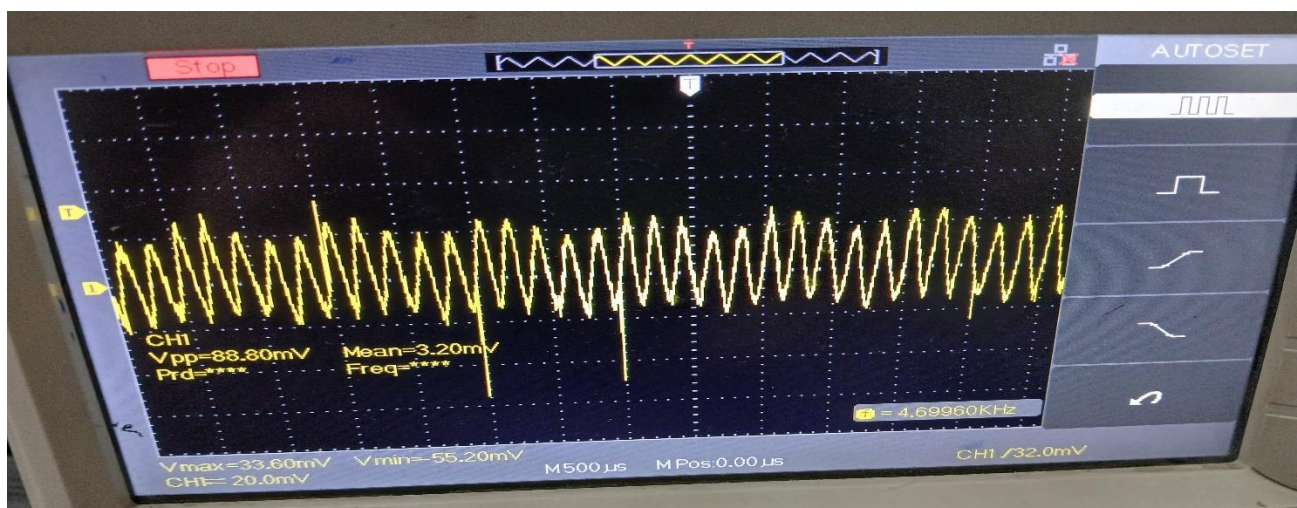


Output 4:

Simulated:

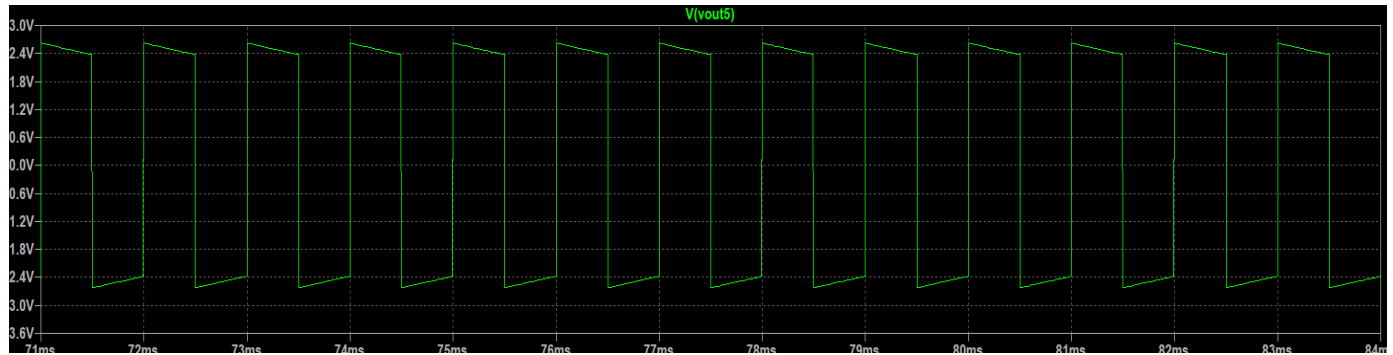


Measured:

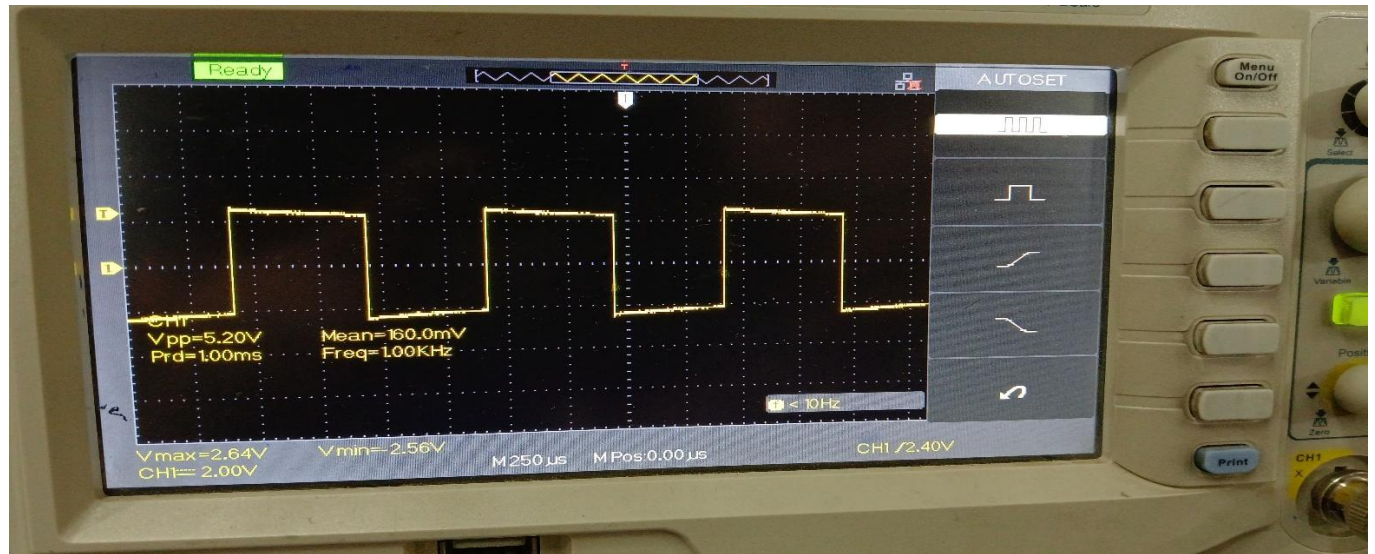


Output 5:

Simulated:



Measured:



Conclusion:

The project is successfully achieved and its objective of extracting and processing various frequency components from a square waveform using a combination of passive and active filter designs. Each filter stage was carefully designed to target specific frequency bands, ensuring accurate signal extraction, amplification, and modulation. Simulation and practical results were in close agreement, validating the theoretical designs. Overall, the system effectively demonstrated core principles of electrical network analysis and signal filtering.